

To change directory as the directory as being created:

```
(bakintunde@kalit04)-[~]
$ cd pixel3

(bakintunde@kalit04)-[~/pixel3]
$ cd Pixel 3
cd: string not in pwd: Pixel

(bakintunde@kalit04)-[~/pixel3]
$ cd "Pixel 3"
1 x

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ wget -q https://corp.digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Cellebrite%20Extraction/Pixel%203.zip

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ wget -q https://digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Cellebrite%20Extraction/Pixel%203.zip

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$
```

Download the image file

```
(bakintunde@kalit04)-[~]
$ cd pixel3

(bakintunde@kalit04)-[~/pixel3]
$ cd Pixel 3
cd: string not in pwd: Pixel

(bakintunde@kalit04)-[~/pixel3]
$ cd "Pixel 3"
1 x

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ wget -q https://corp.digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Cellebrite%20Extraction/Pixel%203.zip

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ wget -q https://digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Cellebrite%20Extraction/Pixel%203.zip

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ ls Pixel\ 3.zip
'Pixel 3.zip'

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ ls Pixel\ 3.zip -l
-rw-r--r-- 1 bakintunde student 5247820897 Nov 21 2020 'Pixel 3.zip'

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$
```

Find the hash value of the file

```

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ hashdeep Pixel\ 3.zip
##### HASHDEEP-1.0
##### size,md5,sha256,filename
## Invoked from: /home/student/pixel3/Pixel 3
## $ hashdeep Pixel 3.zip
##
5247820897,9cc37ebbbc4e918ee5427de1fe1deecc,ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a,/home/student/pixel3/Pixel 3/Pixel 3
.zip

```

Unzip file: file has being unzip before

```

apex  init.environ.rc  init.usb.configfs.rc  init.zygote32.rc  metadata 'Pixel 3'  product  sbin  ueventd.rc
data  init.rc           init.usb.rc           init.zygote64_32.rc  mnt      'Pixel 3.zip'  res      system  vendor

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ unzip -q Pixel\ 3.zip
replace Pixel 3/.DS_Store? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
replace Pixel 3/apex/com.android.apex.cts.shim@1/apex_manifest.json? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
replace Pixel 3/apex/com.android.apex.cts.shim@1/etc/hash.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: A

```

View the list of file in the directory

```

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ ls -l
total 5124940
drwxr-xr-x 16 bakintunde student 4096 Feb 15 2020 apex
drwxr-xr-x 25 bakintunde student 4096 Feb 15 2020 data
-rw-r--r-- 1 bakintunde student 1653 Dec 31 2008 init.environ.rc
-rw-r--r-- 1 bakintunde student 33402 Dec 31 2008 init.rc
-rw-r--r-- 1 bakintunde student 7690 Dec 31 2008 init.usb.configfs.rc
-rw-r--r-- 1 bakintunde student 5649 Dec 31 2008 init.usb.rc
-rw-r--r-- 1 bakintunde student 563 Dec 31 2008 init.zygote32.rc
-rw-r--r-- 1 bakintunde student 981 Dec 31 2008 init.zygote64_32.rc
drwxr-xr-x 4 bakintunde student 4096 Feb 15 2020 metadata
drwxr-xr-x 3 bakintunde student 4096 Feb 15 2020 mnt
drwxr-xr-x 11 bakintunde student 4096 Feb 15 2020 'Pixel 3'
-rw-r--r-- 1 bakintunde student 5247820897 Nov 21 2020 'Pixel 3.zip'
drwxr-xr-x 10 bakintunde student 4096 Feb 15 2020 product
drwxr-xr-x 3 bakintunde student 4096 Feb 15 2020 res
drwxr-xr-x 3 bakintunde student 4096 Feb 15 2020 sbin
drwxr-xr-x 12 bakintunde student 4096 Feb 15 2020 system
-rw-r--r-- 1 bakintunde student 2608 Dec 31 2008 ueventd.rc
drwxr-xr-x 16 bakintunde student 4096 Feb 15 2020 vendor

```

View list of root directory

```

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ ls -l Pixel\ 3
total 104
drwxr-xr-x 16 bakintunde student 4096 Feb 15 2020 apex
drwxr-xr-x 25 bakintunde student 4096 Feb 15 2020 data
-rw-r--r-- 1 bakintunde student 1653 Dec 31 2008 init.environ.rc
-rw-r--r-- 1 bakintunde student 33402 Dec 31 2008 init.rc
-rw-r--r-- 1 bakintunde student 7690 Dec 31 2008 init.usb.configfs.rc
-rw-r--r-- 1 bakintunde student 5649 Dec 31 2008 init.usb.rc
-rw-r--r-- 1 bakintunde student 563 Dec 31 2008 init.zygote32.rc
-rw-r--r-- 1 bakintunde student 981 Dec 31 2008 init.zygote64_32.rc
drwxr-xr-x 4 bakintunde student 4096 Feb 15 2020 metadata
drwxr-xr-x 3 bakintunde student 4096 Feb 15 2020 mnt
drwxr-xr-x 10 bakintunde student 4096 Feb 15 2020 product
drwxr-xr-x 3 bakintunde student 4096 Feb 15 2020 res
drwxr-xr-x 3 bakintunde student 4096 Feb 15 2020 sbin
drwxr-xr-x 12 bakintunde student 4096 Feb 15 2020 system
-rw-r--r-- 1 bakintunde student 2608 Dec 31 2008 ueventd.rc
drwxr-xr-x 16 bakintunde student 4096 Feb 15 2020 vendor

```

To view data in system directory

Command: ls -l "Pixel 3/system"

```
(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ ls -l "Pixel 3/system"
total 60
drwxr-xr-x  2 bakintunde student 4096 Feb 15 2020 apex
drwxr-xr-x  4 bakintunde student 4096 Feb 15 2020 app
drwxr-xr-x  4 bakintunde student 4096 Feb 15 2020 bin
-rw-r--r--  1 bakintunde student 1970 Dec 31 2008 build.prop
drwxr-xr-x 14 bakintunde student 4096 Feb 15 2020 etc
drwxr-xr-x  2 bakintunde student 20480 Feb 15 2020 fonts
drwxr-xr-x  5 bakintunde student 4096 Feb 15 2020 framework
drwxr-xr-x  2 bakintunde student 4096 Feb 15 2020 lib
drwxr-xr-x  2 bakintunde student 4096 Feb 15 2020 lib64
drwxr-xr-x  6 bakintunde student 4096 Feb 15 2020 priv-app
drwxr-xr-x  7 bakintunde student 4096 Feb 15 2020 usr
```

To view items in data directory
Command: ls -l "Pixel 3/data"

```
(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ ls -l "Pixel 3/data"
total 108
drwxr-xr-x  3 bakintunde student 4096 Feb 15 2020 adb
drwxr-xr-x  4 bakintunde student 4096 Feb 15 2020 apex
drwxr-xr-x 83 bakintunde student 12288 Feb 15 2020 app
drwxr-xr-x  3 bakintunde student 4096 Feb 15 2020 backup
drwxr-xr-x  2 bakintunde student 4096 Feb 15 2020 cache
drwxr-xr-x  4 bakintunde student 4096 Feb 15 2020 dalvik-cache
drwxr-xr-x 103 bakintunde student 4096 Feb 15 2020 data
drwxr-xr-x  3 bakintunde student 4096 Feb 15 2020 drm
drwxr-xr-x  2 bakintunde student 4096 Feb 15 2020 magisk_backup_e132ca5af8732354d2f63dcfcd4331cddc309ae8
drwxr-xr-x  3 bakintunde student 4096 Feb 15 2020 media
drwxr-xr-x 20 bakintunde student 4096 Feb 15 2020 misc
drwxr-xr-x  3 bakintunde student 4096 Feb 15 2020 misc_ce
drwxr-xr-x  2 bakintunde student 4096 Feb 15 2020 nfc
drwxr-xr-x  2 bakintunde student 4096 Feb 15 2020 property
drwxr-xr-x  2 bakintunde student 12288 Feb 15 2020 resource-cache
drwxr-xr-x 14 bakintunde student 4096 Feb 15 2020 system
drwxr-xr-x  3 bakintunde student 4096 Feb 15 2020 system_ce
drwxr-xr-x  4 bakintunde student 4096 Feb 15 2020 system_de
drwxr-xr-x  2 bakintunde student 4096 Feb 15 2020 tombstones
drwxr-xr-x  3 bakintunde student 4096 Feb 15 2020 unencrypted
drwxr-xr-x  4 bakintunde student 4096 Feb 15 2020 user_de
drwxr-xr-x 12 bakintunde student 4096 Feb 15 2020 vendor
drwxr-xr-x  3 bakintunde student 4096 Feb 15 2020 vendor_ce
```

To view user_de directory

```
(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ ls -l "Pixel 3/data/user_de/"
total 8
drwxr-xr-x 80 bakintunde student 4096 Feb 15 2020 0
drwxr-xr-x 37 bakintunde student 4096 Feb 15 2020 11
```

To view the items in data/data

```

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ ls 'Pixel 3/data/data/'
com.android.chrome
com.android.hotwordenrollment.okgoogle
com.android.hotwordenrollment.xgoogle
com.android.keychain
com.android.mtp
com.android.nfc
com.android.providers.calendar
com.android.providers.contacts
com.android.providers.downloads
com.android.providers.media
com.android.providers.telephony
com.android.providers.userdictionary
com.android.service.ims.presence
com.android.traceur
com.android.vending
com.discord
com.enflick.android.TextNow
com.facebook.orca
com.google.android.apps.chromecast.app
com.google.android.apps.docs
com.google.android.apps.docs.editors.docs
com.google.android.apps.dreamliner
com.google.android.apps.enterprise.dmagent
com.google.android.apps.gcs
com.google.android.apps.helpertc
com.google.android.apps.magazines
com.google.android.apps.maps
com.google.android.apps.messaging
com.google.android.apps.nexuslauncher
com.google.android.apps.turbo
com.google.android.apps.tycho
com.google.android.apps.walletnfcrel
com.google.android.apps.wallpaper
com.google.android.apps.wellbeing
com.google.android.apps.work.oobconfig
com.google.android.apps.youtube.music
com.google.android.as
com.google.android.calendar
com.google.android.configupdater
com.google.android.contacts
com.google.android.deskclock
com.google.android.dialer
com.google.android.documentsui
com.google.android.ext.services
com.google.android.gm
com.google.android.gms
com.google.android.GoogleCamera
com.google.android.googlequicksearchbox
com.google.android.gsf
com.google.android.ims
com.google.android.inputmethod.latin
com.google.android.keep
com.google.android.markup
com.google.android.music
com.google.android.onetimeinitializer
com.google.android.partnersetup
com.google.android.pixel.setupwizard
com.google.android.projection.gearhead
com.google.android.webview
com.google.android.wfcactivation
com.google.android.youtube
com.google.ar.core
com.google.intelligence.sense
com.google.vr.apps.ornament
com.google.vr.vrcore
com.imgur.mobile
com.imo.android.imous
com.instagram.android
com.instagram.threadsapp
com.mewe
com.mywicker.wicker2
com.qti.ltebc
com.radicalapps.cyberdust
com.silentcircle.silentphone
com.skout.android
com.skype.raider
com.snapchat.android
com.spotify.music
com.thinkyeah.galleryvault
com.topjohnwu.magisk
com.twitter.android
com.venmo
com.viber.voip
com.whatsapp
com.wire
com.zhiliaoapp.musically
jp.naver.line.android

```

This shows that Twitter was replaced X Corp on the website

```

<div class="F93Bb ynrBgc xwcR9d">...
  </div>
  <div class="tv4jIf">
    <div class="Vbfug auoI0c">
      <a href="/store/apps/developer?id=X+Corp">
        <span>X Corp.</span> == $0
      </a>
    </div>
  </div>
  <div class="ulKokd">...</div>
</div>

```

The shows the update command instead of Twitter its now X Corp

```

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ curl --no-progress-meter https://play.google.com/store/apps/details?id=X+Corp. |grep -Pio 'itemprop="name"><span>X Corp.</span>'

```

To view the apps with money transfer function:

```

(bakintunde@kalit04)-[~/pixel3/Pixel 3]
$ ls 'Pixel 3/data/data' \
| while read package; do \
curl --no-progress-meter https://play.google.com/store/apps/details?id=$package \
| grep -Pio 'itemprop="name"><span>\K.*?(?=</span>)' \
&& echo $package; \
done

```

```

$ ls 'Pixel 3/data/data' \
| while read package; do \
curl --no-progress-meter https://play.google.com/store/apps/details?id=$package \
| grep -Pio '<div jsname="sngbd">\K.*?(?=</div>)' | grep -io money\
&& echo $package; \
done
money
money

```